



3D Modelling of G+1 Residential Building

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Outline

- Introduction to AutoCAD: A Brief Overview
- Project Brief: The G+1 Residential Building
- Phase 1: 2D Drafting (The Blueprint)
- Phase 2: 3D Modeling (Bringing it to Life)
- Conclusion

About AutoCAD

- **What is it?** A commercial Computer-Aided Design (CAD) software for 2D and 3D drafting.
- **Who uses it?** Architects, Engineers, Project Managers, and other design professionals.
- **Why is it an industry standard?** It offers unparalleled precision, efficiency for edits, and collaborative power through its standard **.DWG** file format.



2D Commands

These commands were used to create the basic geometry of the floor plans.

- **UNITS & LIMITS:** To set up the drawing environment.
- **LINE / POLYLINE:** To draw wall outlines. Polyline is key for creating closed shapes.
- **RECTANGLE:** For creating columns and basic room shapes.
- **CIRCLE & ARC:** For features like door swings.

These commands were used to refine the geometry into a clean plan.

- **OFFSET:** The most important command for creating wall thickness.
- **TRIM & EXTEND:** Essential for cleaning up wall intersections.
- **MOVE & COPY:** For placing and duplicating objects like columns.
- **MIRROR:** To create symmetrical parts of the building plan efficiently.

To make the plan readable, professional, and easy to manage.

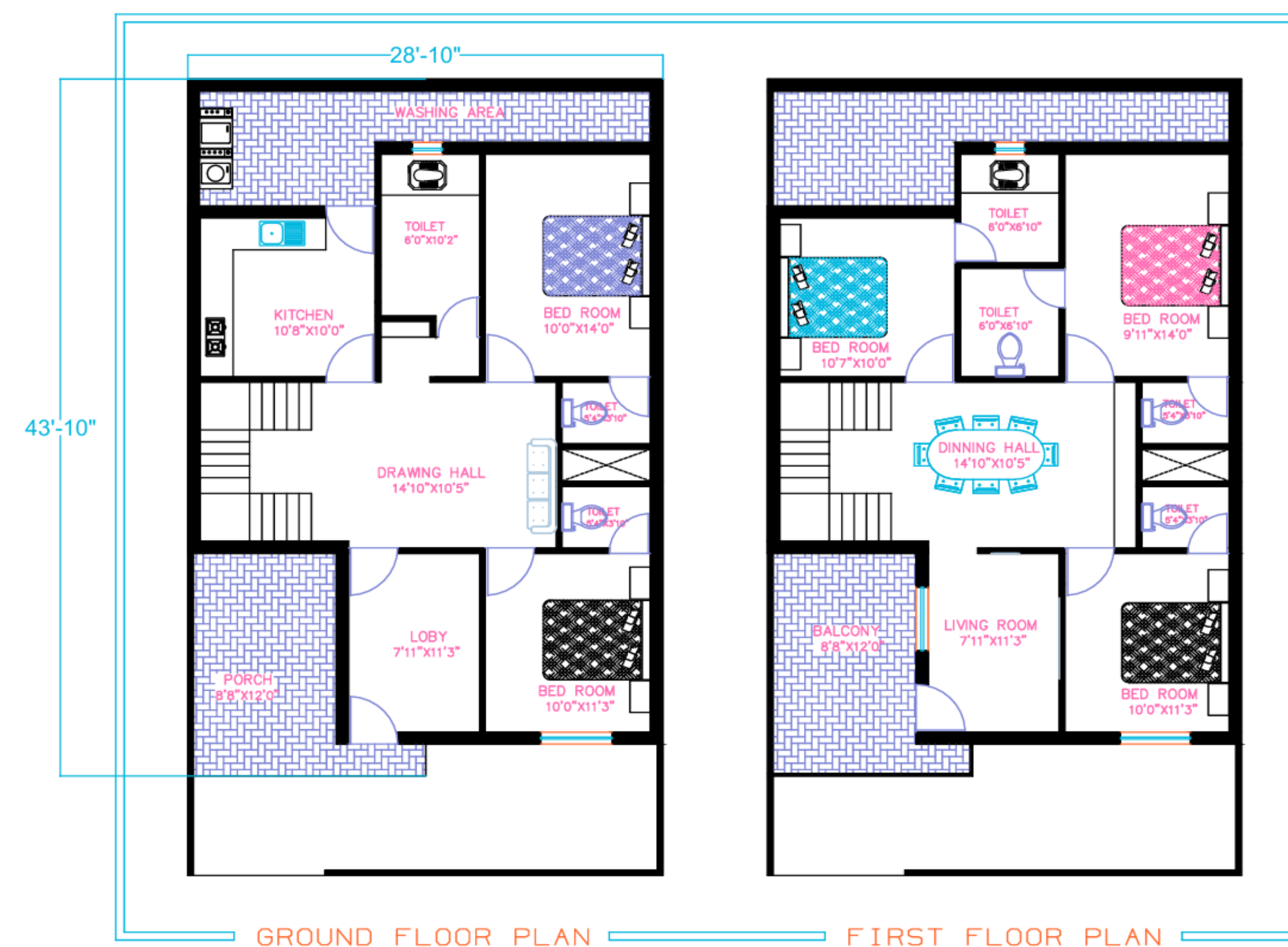
- **LAYER:** The most critical tool. I used separate layers for Walls, Doors, Windows, Text, Dimensions, etc.

- **BLOCK / INSERT:** To create reusable symbols for doors, windows, and furniture.
- **HATCH:** To fill walls with a solid or patterned fill for clarity.
- **DIMSTYLE / DIMENSION:** To add all necessary measurements.

2D Drafting

PLAN:

BUILT UP AREA 29'X44'(IN BETWEEN)



ELEVATION:



3D Commands

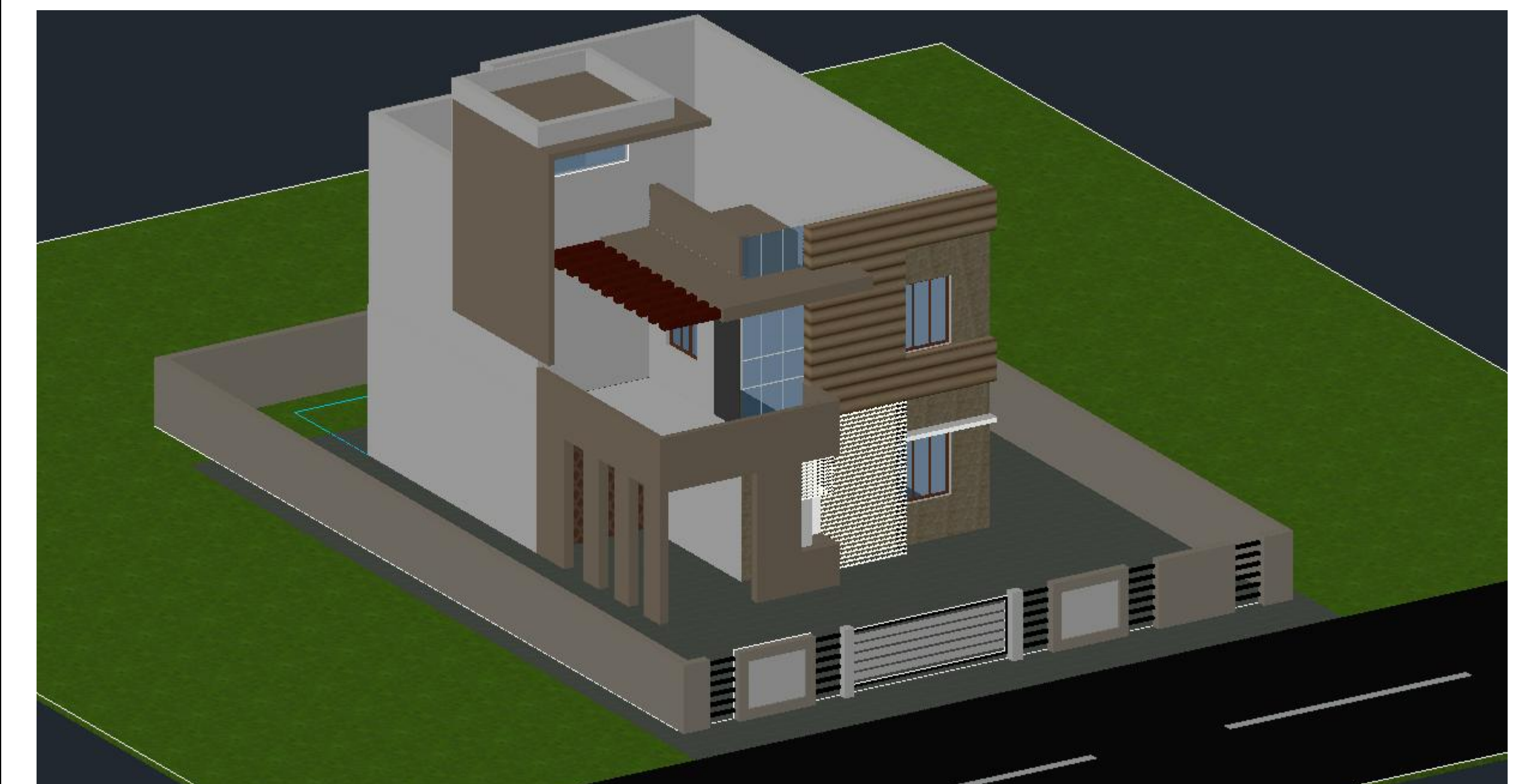
These commands were used to convert the 2D linework into 3D solids.

- **EXTRUDE:** The primary command to give height to wall outlines.
- **PRESSPULL:** An intuitive way to raise areas and create slabs or walls.

Boolean Operations:

- **UNION:** To join wall and column solids together.
- **SUBTRACT:** Crucial for creating openings for doors and windows.
- **SWEEP:** Used to create complex objects like stairway handrails.

3D Modelling



Rendered Image:



Conclusions

- We successfully moved from a 2D concept to a 3D visualization, all within AutoCAD.
- The 2D phase ensured technical accuracy and planning.
- The 3D phase provided a clear, realistic representation for design validation and client presentation.
- This workflow highlights AutoCAD's power as a comprehensive tool for architectural design.